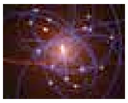
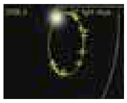


Galactic Centre Map
These series of animations by UCLA track the motions of stars around Sagittarius A*. **https://dgit.in/apr20-03**



S2 and Sgr A*
An interview with astronomer Stefan Gillessen on S2 star and what we know about Saggitarius A*. **https://dgit.in/apr20-04**

compact object in the middle of it all, designated Sagittarius A*. Around Sagittarius A* were around 40 other stars in close orbits, but in various orbital planes unlike the planets in the Solar System. It was only after decades of observations using a series of increasingly powerful and sophisticated radio telescopes, that astronomers became confident that the Milky Way was housing a supermassive black hole at its core, and that supermassive black holes existed at all. The European Southern Observatory and the Max Planck Institute for Extraterrestrial Physics were both responsible for these observations. However, the scientific method requires taking nothing for granted and an abundance of caution, so the object is still at times referred to as a supermassive black hole candidate in scientific literature, although the GRAVITY instrument removed ambiguity in the matter in 2018 itself.

Sagittarius A* is one of the primary targets of interest for the GRAVITY instrument. In fact, the location of the observatory was selected because the heart of the Milky Way appears in the southern hemisphere. The GRAVITY instrument will be able to probe the gravitational field around Sagittarius A* in detail, observing the movements of the stars closest to it. The GRAVITY instrument went online just in time to observe one of the stars in the galactic centre, S2, at its closest approach to Sagittarius A*, at a distance of 17 light hours from the supermassive black hole, moving at speeds close to 3 percent that of light.

VERIFYING EINSTEIN'S PREDICTIONS

During the observation, the team combined the information from GRAVITY, which provided accurate measurements of distances, and the measurement from another VLT instrument known as SINFONI (Spectrograph for INtegral Field Observations in the Near Infrared). The observations were compared to predictions made by both Newtonian physics, and Einstein's Theory of Relativity. The trajectory of S2 followed the predictions by Einstein's Theory of Relativity, but not the Newtonian physics.

One of the predictions by Einstein's theory was what is known as gravitational redshift. This means that under the strong gravitational influence of another object, the colour of a star can shift towards the red portions of the spectrum. This is precisely what occurred to S2, as it made its closest approach to Sagittarius A* which occurs twice during its sixteen year long orbit. The title image of this article is an artistic illustration of this red shift. The VLT and Gravity instrument continue to observe S2, in the hopes of observing another effect related to Einstein's theories. The orbital plane of the star is expected to shift slightly because of a relativistic effect called Schwarzschild precession. The path of the star around such a massive object is somewhat akin to the path that the pen tip takes in a spirograph. Testing these theories within the Solar System or in a laboratory is simply not possible because of the extreme gravitational fields involved. Such observations can only be conducted directly, and the closest object we have with exotic physics on the fringes is the one at the center of our own galaxy.

Credit: ESO/L Calcada

An illustration of HR8799e, the first exoplanet observed by the GRAVITY instrument.

The GRAVITY instrument was subsequently used to track the movement of material very close to the event horizon of Sagittarius A*, clumps of gas moving at speeds close to 30 percent that as the speed of light. GRAVITY also observed infrared flares from this accretion disk in 2018, which confirmed that Sagittarius A* was in fact, a supermassive black hole. Three flares were observed, closely matching the predictions of the hotspots in orbit around a black hole as massive as Sagittarius A*.

CHARACTERISING EXOPLANETS

In 2019, the GRAVITY instrument was used to observe an exoplanet, HR8799e, that is 129 light years away from the Earth towards the constellation of Pegasus. The complex series of mirrors, polarising filters and optical fibers guiding the light was used to accurately split up the light coming from the planet, and the light coming from the Star. This observation was the first time that such a disentanglement was endeavoured on a body, and allowed scientists to peer into the very atmosphere of this distant planet, and explore its chemical composition.

The planet is a Super Jupiter, a gas giant that is orbiting close to the host star. Our Solar System does not have any comparable objects. The planet is young in astronomical timescales, at only 30 million years. It is also searing hot, with temperatures at around 1000 °C. In fact, the scientists believe that they may be have witnessed the birth pangs of a giant exoplanet. This is because the gas giant seemed to have a source of illumination in its interior, with rays of warm light coming up from the depths, and breaking through the clouds above. The clouds themselves were made up of silicon and iron compounds, swirling around the plant in a colossal storm system. The finding indicates that the planet may be hot enough to see a hot rain of silicon and iron. The observation demonstrated that the GRAVITY instrument is a new way of characterising exoplanets. d